



StockMind – Inventory & Warehouse Management

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2026-27

Abstract

- StockMind is a web app to track stock, suppliers and orders.
- Main Outcome: fewer stock-outs, less manual work.

Problem Statement & Objectives

- Problem: Manual stock tracking causes errors and stock-outs.
- Objective 1: Track products and stock levels.
- Objective 2: Record purchase and sales orders.
- Objective 3: Auto low-stock alerts and reports.
- Scope: Admin and staff; stock, orders, reports.

Existing vs Proposed System

Feature	Existing System	Proposed System
Tracking	Excel sheets	Live Database
Orders	Paper records	Digital Orders
Alerts	Checked by hand	Auto Alerts
Reports	Slow, manual	Instant Reports
Limitation	Human errors	Central Database

Feasibility Study

- Technical: React, Node.js and MySQL – widely used, well documented tools.
- Economic: Open-source stack and free-tier hosting.
- Operational: Simple screens, easy for admins to learn.
- Schedule: Completion within the semester timeline.

Results & Testing

- All CRUD operations work correctly.
- Validation and low-stock alerts tested.

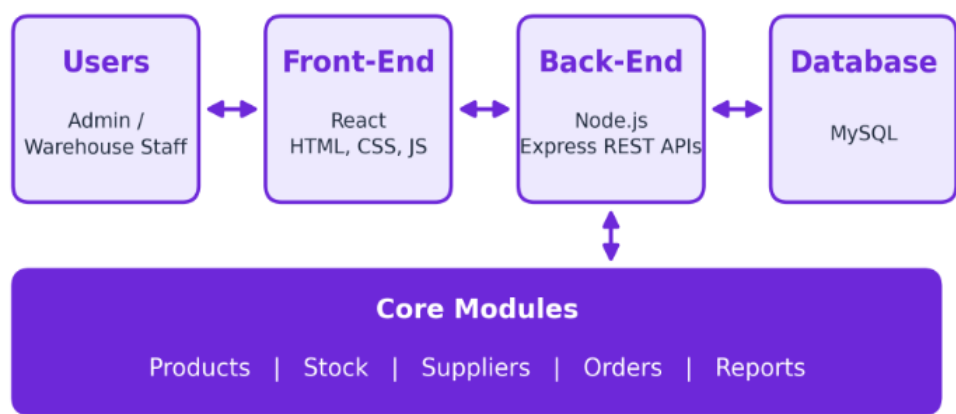
Conclusion

- StockMind simplifies warehouse work.
- It achieved all 3 objectives.

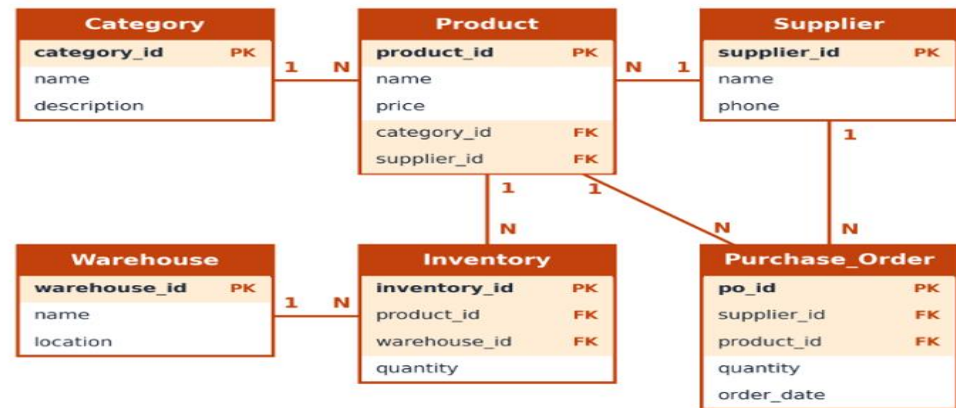
Tools & Technologies

- Front-End: React , HTML, CSS, JavaScript.
- Back-End: Node.js + Express REST APIs.
- Database: MySQL.
- Tools: Git & GitHub, Postman, VS Code.

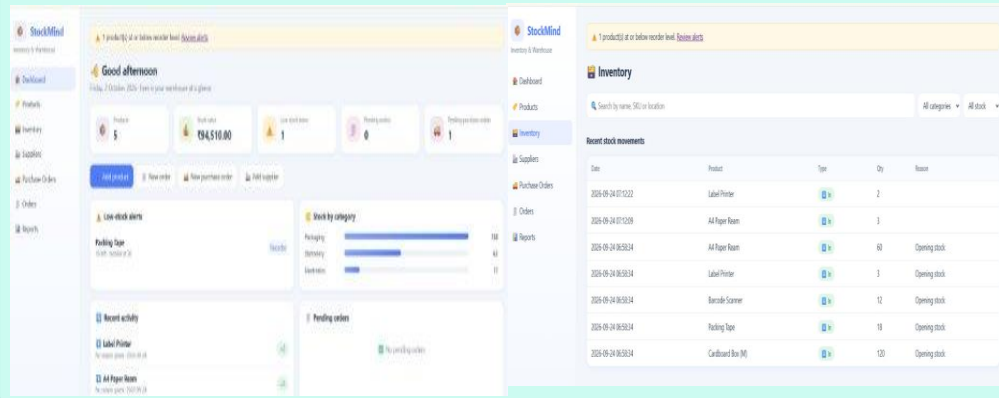
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Barcode / QR code scanning.
- Mobile app and demand forecasting.